

submission-draft.md

Digital Health Systems Optimization Challenge

Executive Summary

This submission redesigns the **high-risk pregnancy referral and continuity workflow** in India's public health system. The core problem is not simply detecting high-risk pregnancies, but converting detection into **timely, complete, and closed-loop referrals** across rural clinics, district hospitals, and follow-up care. The proposed digital workflow uses structured screening, a minimum referral dataset, transport coordination, interoperable handoff, and closed-loop follow-up to reduce avoidable delay, duplicate work, and loss to care.

1. Healthcare Workflow Analysis

Current-state summary

The present workflow depends heavily on paper records, manual communication, and disconnected facility processes. This creates delays at referral decision, transport coordination, district intake, and follow-up closure.

Main inefficiencies

- Late or inconsistent risk screening
- Incomplete referral packets
- Transfer delays with weak visibility
- Duplicate registration and tests at district level
- Poor feedback to origin clinic
- Missed follow-up that remains invisible

Why this matters

India's maternal mortality ratio for 2018-20 is reported at **97 per 100,000 live births**. Maternal deaths are largely preventable when women receive timely skilled care, emergency management, and continuity across levels of care.

2. Digital Workflow Redesign

Proposed future-state design

1. Structured ANC risk capture at the primary facility.
2. Rules-based referral prompts with clinician confirmation.
3. Auto-generated digital referral packet with minimum dataset.
4. Live transport/destination status updates.
5. Pre-registration and shared summary at district intake.
6. Digital discharge/follow-up return notification to origin clinic.
7. Reminder and escalation system for missed follow-up.

Digital levers

- Automation
- Decision support
- Interoperability
- Tracking and alerts
- Offline-first workflow

Design safeguards

- Clinician remains final decision-maker.
- Paper fallback remains available.
- Alerts are threshold-based to limit fatigue.
- All claims without verified local data are labeled as estimates.

3. Impact Assessment

Expected outcomes

- Faster referral processing
- Higher referral packet completeness
- Lower duplicate intake and documentation

- Better visibility of transfer and follow-up status
- Reduced missed follow-up

Scenario targets

- Referral packet completeness > 90%
- Closed-loop referral closure > 95%
- District awareness within 15 minutes of referral
- 30%+ reduction in missed follow-up

Credibility and assumptions

All numeric assumptions are tagged as observed, estimated, or scenario target using the source ledger. Source-backed values are drawn from WHO, UNICEF, NHM/ABDM context, and related public-health references. Any local-facility metric not directly sourced should be treated as “needs local validation.”

4. Risks and Mitigations

- Alert fatigue -> limit alerts to critical thresholds.
- Connectivity issues -> offline capture and later sync.
- Adoption gaps -> pilot first, then scale.
- Data quality issues -> mandatory fields and validation.
- Interoperability limitations -> lightweight QR/PDF fallback.

5. Conclusion

The most meaningful gain comes from building a **closed-loop referral operating system** for maternal care. This is realistic, high-impact, and directly aligned with preventable maternal mortality reduction.

stage1-understand.md

Stage 1 - Understand

Workflow Selected

****High-risk pregnancy referral and continuity workflow**** in India's public health system:

****Rural/Primary ANC point -> District hospital -> Specialist follow-up -> Return continuity at primary level****

1. Scenario Context and Goal

The workflow addresses preventable maternal and neonatal harm caused by delayed risk detection, delayed referral transfer, fragmented handoffs, and weak post-referral continuity.

****Goal:**** ensure that high-risk pregnancies are identified early, referred on time with complete information, managed quickly at higher centers, and followed up without loss to care.

2. Scope Boundaries (No Ambiguity)

Included in scope

1. First ANC registration through risk stratification at primary/rural level.
2. Referral decision and referral packet preparation.
3. Transport coordination and district-hospital arrival.
4. District triage, specialist consult, stabilization/admission/discharge decision.
5. Follow-up scheduling and continuity loop back to primary facility.

Excluded from scope

1. Long-term neonatal care after postnatal discharge.
2. Private tertiary-hospital financial workflows.
3. National policy reform beyond facility/network operations.
4. Drug supply-chain redesign outside referral workflow touchpoints.

3. Stakeholders and Roles

| Stakeholder | Current role | Current pain exposure |

| ---|---|---|

| Pregnant woman + caregiver | Seeks ANC, acts on referral advice, navigates transfer/follow-up | Delay, confusion, out-of-pocket travel burden, repeated tests |

| ASHA/ANM/staff nurse (primary) | ANC intake, screening, counseling, referral initiation | Manual forms, protocol variability, weak referral visibility |

| Primary medical officer | Confirms risk severity and referral need | Incomplete prior records, pressure under uncertainty |

| Ambulance/transport coordinator | Arranges patient transfer | Poor coordination and status communication gaps |

| District triage + OB-GYN team | Receives and manages referred case | Incomplete handoff data, queue pressure |

| HMIS/data entry personnel | Enters program/reporting data | Retrospective entry lag and data-quality inconsistencies |

| Program manager | Monitors performance and quality indicators | Limited real-time referral intelligence |

4. Current-State Journey (As-Is)

1. **ANC registration at rural clinic**

- Demographics and obstetric history documented in registers/cards.
- Vitals/basic labs captured variably by site capacity.

2. **Risk screening and categorization**

- Provider screens for danger signs/risk factors.
- Risk marking may be non-standard when workload is high.

3. **Referral decision**

- If high-risk or complication suspected, provider initiates referral.
- Clinical note and referral slip are prepared manually.

4. ****Transport and transfer****

- Ambulance/transport is requested; availability and response vary.
- Family often manages interim communication between facilities.

5. ****District hospital intake****

- Patient arrives; triage and registration repeated.
- Missing referral details can trigger duplicate evaluation/tests.

6. ****Specialist management****

- Case reviewed by OB-GYN team; admitted, stabilized, or discharged.
- Plan may not flow back cleanly to primary clinic.

7. ****Post-referral follow-up****

- Follow-up dates given; adherence depends on counseling and logistics.
- Primary facility may not have timely visibility on completed care.

5. Key Constraints

1. Intermittent connectivity and uneven digital maturity across facilities.
2. High workforce load and time-limited consultations.
3. Paper-heavy handoffs and fragmented records across care levels.
4. Transport, bed, and specialist availability uncertainty.
5. Limited closed-loop tracking from referral origin to follow-up completion.

6. Baseline Evidence Anchors (Credible)

1. India MMR has declined, reported as ****97 per 100,000 live births (2018-20 period)****.

- Label: ****Observed**** (Source ledger C1; primary citation S1, cross-reference S2)

2. Maternal deaths are largely preventable with timely skilled care and emergency management.

- Label: ****Observed**** (C2; S2/S3)

3. Major direct causes include hemorrhage, sepsis/infection, hypertensive disorders, delivery complications, unsafe abortion.

- Label: ****Observed**** (C3; S2/S3)

7. Explicit Assumptions (Frozen for Next Stages)

1. Facilities in scope are public-sector primary units linked to a district hospital.

2. At least one ambulance pathway exists but response and coordination are inconsistent.

3. Primary and district sites do not share a fully interoperable longitudinal maternal record today.

4. Digital redesign must function with low-bandwidth fallback and partial offline operation.

5. Any metric without verified local baseline will use ****estimated range**** from the source ledger and be marked “needs local validation.”

8. Problem Statement for Stage 2

The core system problem is not only identifying high-risk pregnancies, but ensuring ****time-critical, information-complete, closed-loop referrals**** with reliable follow-up continuity across fragmented service layers.

stage2-analyze.md

Stage 2 - Analyze

1. What Is Going Wrong Today

The current referral pathway fails less because high-risk pregnancies are unidentified and more because the system cannot reliably convert risk recognition into **timely, complete, and closed-loop action**.

Core bottlenecks

1. **Delayed risk recognition**

- Risk screening may happen late or inconsistently.
- Danger signs are not always standardized across busy primary clinics.

2. **Referral information fragmentation**

- Referral slips may omit key obstetric history, vitals, labs, or reason for referral.
- District teams often repeat assessment/tests because the original packet is incomplete.

3. **Transport coordination lag**

- Ambulance or transfer arrangement is not always visible in real time.
- Family and staff spend time on calls and manual follow-up.

4. **District intake duplication**

- Patient gets re-registered and re-triaged.
- Specialist review is slowed by queue pressure and missing clinical context.

5. **Weak post-referral continuity**

- Primary site may not know whether the patient arrived, was admitted, or needs return follow-up.
- Missed follow-up is often invisible until a complication occurs.

6. **Data latency and poor observability**

- HMIS data is often retrospective, so managers cannot intervene in real time.
- Operational blind spots remain hidden until audit time.

2. Root-Cause Analysis

| Problem | Likely root cause | Category | Why it matters |

| --- | --- | --- | --- |

| Late or inconsistent risk detection | Non-standard screening workflow, high workload, limited decision support | Process / People | Delays referral and increases emergency presentations |

| Incomplete referral packet | Paper-only handoff, no minimum dataset, time pressure | Data / Process | Forces duplicate assessments and slows care |

| Transfer delay | No live transport status, weak escalation channel | Coordination / System | Time-sensitive cases worsen during waiting |

| Repeated intake at district | Records not shared, no interoperable patient identity | Data / Interoperability | Wastes time and staff capacity |

| Missed follow-up | No closed-loop tracking and reminder pathway | Process / System | Continuity leakage, preventable readmissions/complications |

| Limited manager visibility | Manual reporting lag, no operational dashboard | Data / Governance | Problems are discovered too late to correct |

3. Effects by Domain

Patient experience

- More waiting, uncertainty, and repeated explanations.
- Higher travel burden and anxiety during already high-risk pregnancies.
- Lower trust when referral instructions are unclear or inconsistent.

Clinical and operational efficiency

- More duplicate registrations, duplicate tests, and rework.
- Triage and specialist time consumed by avoidable administrative gaps.
- Ambulance and bed capacity are used less efficiently because demand is not visible early.

Decision quality and resource use

- Care teams make decisions with incomplete information.
- High-risk cases may be under-triaged or escalated too late.
- Resources are spent on rework instead of active management of time-critical cases.

4. Prioritized Issues

Priority	Issue	Impact	Feasibility	Why prioritized
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1	Incomplete referral handoff	Very high	High	Directly slows treatment and causes repeat work
2	No closed-loop referral tracking	Very high	High	Main cause of continuity loss and invisible failures
3	Transport coordination lag	High	Medium	Strong effect on outcomes, moderate digitization effort
4	Late/inconsistent risk screening	High	Medium	Prevents timely referral at source
5	District intake duplication	Medium-High	Medium	Efficiency loss that can be reduced with interoperability

5. Design Requirements Derived From Analysis

1. A minimum referral dataset must be mandatory for every high-risk transfer.
2. Referral status must be visible across levels (sent, accepted, arrived, seen, discharged, follow-up due).
3. The system must support offline capture and sync later to avoid connectivity failure.
4. The original clinic must receive outcome feedback automatically.

5. Low-friction decision support must assist, not replace, clinical judgment.

6. Evidence Anchors

- India MMR for 2018-20: **97** per 100,000 live births (**Observed**).
- WHO ANC guidance emphasizes person-centred, quality ANC with assessment and interventions that can be operationalized in digital systems (**Observed**).
- WHO intrapartum guidance emphasizes evidence-based, woman-centred care and adaptable local protocols (**Observed**).
- ABCD/ABDM-style interoperability is a realistic digital backbone for shared records (**Observed** from mission context, but design use is a **Scenario target** until local deployment details are verified).

7. Analysis Summary

The biggest leverage is not inventing a new clinical model; it is building a **closed-loop referral operating system** that removes information loss, reduces delays, and makes transfers visible in real time.

stage3-design.md

Stage 3 - Design

Future-State Workflow

The redesigned model is a **closed-loop maternal referral workflow** that connects primary ANC sites, district hospitals, and follow-up care through a shared digital referral process.

1. Design Principles

1. **Clinically led**: supports decision-making; does not replace clinicians.
2. **Low-bandwidth resilient**: works offline and syncs later.
3. **Minimum dataset first**: no referral leaves without key fields.
4. **Closed-loop by default**: every referral must end in a visible outcome.
5. **Patient-centered**: reduces repeated explanation, travel uncertainty, and follow-up leakage.

2. Future-State Flow

Step 1: Structured ANC risk capture

- ASHA/ANM/staff nurse records pregnancy details in a digital ANC form.
- System prompts for red flags and risk factors using a standardized checklist.
- If danger signs are present, the case is auto-tagged as high priority.

Step 2: Decision support and referral trigger

- A rules-based alert recommends referral when defined thresholds are met.
- The clinician confirms or overrides the recommendation with a mandatory reason.
- This preserves judgment but reduces missed escalation.

Step 3: Digital referral packet generation

- The system creates a referral packet containing:

- patient identity
- gestational age
- vitals
- danger signs
- relevant history
- lab results if available
- reason for referral
- urgency level
- origin facility
- Packet is available as QR/PDF/offline sync format.

Step 4: Transport and destination coordination

- Referral is broadcast to district hospital queue and transport coordinator.
- Status updates show whether ambulance is requested, assigned, en route, or arrived.
- If transport is delayed beyond threshold, escalation alert is triggered.

Step 5: District intake and triage

- Destination hospital scans or opens the packet.
- Patient is pre-registered and triaged using prior data.
- Specialist sees the same referral summary rather than re-entering basics.

Step 6: Specialist management and discharge planning

- OB-GYN team records management outcome:
 - admitted
 - treated and discharged
 - referred onward
 - follow-up required

- Discharge/follow-up instructions are returned digitally to origin clinic and patient-facing channel.

Step 7: Closed-loop follow-up

- Primary clinic receives automatic notification of outcome and due date.
- Missed follow-up triggers reminders to the care team and community worker.
- High-risk unresolved cases remain on an exception list until closed.

3. Digital Levers Used

Digital lever	How it is used	Problem solved
Automation	Risk flagging, packet creation, reminders	Manual delay and omissions
Decision support	Referral threshold prompts	Inconsistent risk recognition
Interoperability	Shared referral summary across levels	Duplicate registration and data loss
Tracking/alerts	Transport status and follow-up monitoring	Invisible referral failures
Offline-first design	Local capture with later sync	Connectivity gaps

4. Role-Based Actions

Role	Digital action	Exception handling
ANM/ASHA	Enters ANC data, initiates referral	Offline capture allowed; sync later
Medical officer	Confirms referral and urgency	Must record reason if overriding alert
Transport coordinator	Accepts and updates transfer status	Escalation if threshold exceeded
District triage nurse	Reviews referral packet	If packet incomplete, request missing item but do not restart case

| OB-GYN specialist | Records outcome and follow-up plan | Mandatory discharge outcome entry |

| Primary clinic staff | Receives outcome and follows up | Escalate if patient misses return visit |

5. Mapping Design to Priority Problems

| Priority problem | Design response |

|---|---|

| Incomplete referral handoff | Mandatory minimum dataset + auto-generated referral packet |

| No closed-loop tracking | Status lifecycle + outcome notification + exception queue |

| Transport coordination lag | Shared transfer board + escalation alerts |

| Late/inconsistent risk screening | Embedded checklist + rules-based prompts |

| District duplication | Shared summary + pre-registration + QR/packet reuse |

6. Assumptions and Guardrails

1. The district hospital is willing to accept pre-notified referrals.
2. The system can integrate at least minimally with digital records or operate as a standalone overlay.
3. Paper fallback remains available for outages.
4. The design does not require full AI diagnosis; only rules-based support.
5. Alerts are rate-limited to prevent fatigue.

7. Scenario Target Outcomes

- Referral packet completeness: **target > 90%**.
- Closed-loop referral closure: **target > 95%**.
- Time from referral decision to district awareness: **target < 15 minutes**.
- Missed follow-up rate: **target reduction by 30%+ from baseline**.

All scenario targets are ****not observed data****; they are design objectives for Stage 4 impact modeling.

stage4-impact.md

Stage 4 - Impact Assessment

1. Measurement Framework

The redesign will be evaluated using operational, clinical-process, and continuity metrics.

Core KPIs

KPI	Baseline reference	Target	Type
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Referral packet completeness	40% - 70%	> 90%	Estimated baseline / Scenario target
Referral decision-to-district awareness time	60 - 180 min	< 15 min	Estimated baseline / Scenario target
Referral decision-to-arrival time	105 - 360 min	60% reduction from baseline	Estimated baseline / Scenario target
Closed-loop referral closure rate	20% - 50%	> 95%	Estimated baseline / Scenario target
Missed follow-up within 14 days	20% - 45%	30% reduction from baseline	Estimated baseline / Scenario target
Duplicate documentation / re-entry	15% - 35%	50% reduction from baseline	Estimated baseline / Scenario target

2. Impact Logic

A. Reduced waiting and transfer delay

Why it improves:

- Referral packet is ready instantly.
- Transport can be alerted earlier.
- District hospital can prepare before arrival.

Expected effect:

- Lower decision-to-departure time.
- Lower idle waiting at destination.
- Better use of ambulance and triage resources.

B. Better workflow efficiency

Why it improves:

- District staff avoid re-entering the same information.
- Primary staff no longer chase status manually.
- Managers can see bottlenecks earlier.

Expected effect:

- More cases handled per shift without increasing workload linearly.
- Less duplication and less administrative drag.

C. Better patient experience and decision quality

Why it improves:

- Women and families receive clearer referral instructions.
- Specialists see a more complete clinical picture.
- Follow-up is not left to memory alone.

Expected effect:

- Less uncertainty and fewer repeated explanations.
- Faster, safer decisions for high-risk cases.

3. Simple Impact Model

Example modeling assumptions

- If 1,000 high-risk referrals occur annually:

- 500 are currently incomplete or delayed beyond acceptable thresholds (estimated).
- The redesigned workflow improves packet completeness to 90%+.
- Assume 30% of cases currently experience avoidable delay-related rework.

Conservative modeled effect

1. **Packet completeness gain**

- If baseline is 60% complete and target is 92%, then 320 more complete packets per 1,000 referrals are available for first-contact management.

2. **Follow-up leakage reduction**

- If baseline missed follow-up is 30% and target is 21% (30% relative reduction), 90 additional women per 1,000 avoid being lost to follow-up.

3. **Administrative rework reduction**

- If duplicate documentation affects 25% of referrals and is cut by half, 125 cases per 1,000 avoid duplicate entry.

These are **Estimated** values built from the scenario ranges in the source ledger and must be labeled “needs local validation” in the final submission.

4. Risks and Mitigations

Risk	Why it matters	Mitigation
Alert fatigue	Users ignore prompts	Keep alerts rule-based and thresholded
Partial adoption	Workflow breaks at one facility	Start with pilot network and phased rollout
Poor data quality	Bad input weakens decision support	Use mandatory fields and validation rules
Connectivity issues	Sync delays undermine continuity	Offline capture + later sync

| Interoperability limits | Not all systems can integrate | Use lightweight overlay and QR/PDF fallback |

5. Implementation Sequence

1. Pilot with one district and its linked primary centers.
2. Introduce minimum referral dataset and packet generation.
3. Add transfer status board and follow-up notifications.
4. Connect outcome reporting back to origin clinic.
5. Expand to additional facilities after error and adoption review.

6. Bottom-Line Expected Outcome

The redesign should reduce avoidable delay, make referrals visible, and improve continuity without requiring a large-scale AI or infrastructure overhaul. The main value comes from ****information continuity + coordination + accountability****.

source-ledger.md

Source Ledger - High-Risk Pregnancy Referral & Continuity (India Context)

This ledger tracks evidence used in the challenge deliverables and labels each numeric/analytical claim as:

- **Observed**: direct source-backed value
- **Estimated**: computed/assumed for scenario modeling
- **Scenario target**: design objective for future-state performance

A. Core Sources (Credibility First)

| ID | Source title | Organization / Publisher | Year | URL | Reliability tier | Primary use |

| --- | --- | --- | --- | --- | --- | --- |

| S1 | Maternal Mortality in India: Trends and Current Status (SRS Special Bulletin) | Office of Registrar General, India | 2022 (data period 2018-20) | <https://censusindia.gov.in> | A | National MMR baseline and trend |

| S2 | UNICEF India - Maternal Health | UNICEF India | 2024 (page update context) | <https://www.unicef.org/india/maternal-health-0> | A | India MMR summary and preventable cause framing |

| S3 | Trends in Maternal Mortality 2000 to 2020 | WHO, UNICEF, UNFPA, World Bank, UN DESA | 2023 | <https://www.who.int/publications/i/item/9789240068759> | A | Global/LMIC benchmarking and preventable mortality framing |

| S4 | WHO recommendations on antenatal care for a positive pregnancy experience | WHO | 2016 (still foundational guidance) | <https://www.who.int/publications/i/item/9789241549912> | A | ANC quality, visit content, screening framing |

| S5 | WHO recommendations: Intrapartum care for a positive childbirth experience | WHO | 2018 | <https://www.who.int/publications/i/item/9789241550215> | A | Referral/complication management principles |

| S6 | National Health Mission portal and RMNCH+A program documents | MoHFW / NHM India | Ongoing | <https://nhm.gov.in> | A | Public-system architecture and implementation context |

| S7 | LaQshya - Labour Room Quality Improvement Initiative guidance | MoHFW India | 2017+ | <https://nhm.gov.in> | A | District/labour-room quality and emergency readiness context |

| S8 | Janani Shishu Suraksha Karyakram (JSSK) / Janani Suraksha Yojana (JSY) guidelines | MoHFW India | Various | <https://nhm.gov.in> | A | Referral, transport entitlement, institutional care pathway context |

| S9 | Delays in emergency obstetric care literature (Three Delays framework applications in LMICs) | Peer-reviewed literature | Multiple | (to be cited per specific paper used) | A/B | Root-cause structure for referral delay analysis |

| S10 | Digital health architecture and ABDM resources | NHA / GoI | Ongoing | <https://abdm.gov.in> | A | Interoperability and shared-record design assumptions |

B. Confirmed Observed Claims Used in This Project

| Claim ID | Claim | Type | Source ID | Notes |

| --- | --- | --- | --- | --- |

| C1 | India MMR for 2018-20 reported as **97 per 100,000 live births** | Observed | S2 (and cross-check with S1) | Use S1 as primary citation in final submission |

| C2 | Maternal mortality is largely preventable with timely skilled care and treatment | Observed | S2, S3 | Narrative framing claim |

| C3 | Major direct causes include severe bleeding, infections, hypertensive disorders, delivery complications, unsafe abortion | Observed | S2, S3 | Clinical prioritization context |

C. Safe Baseline Ranges for Scenario Modeling (When Local Facility Data Is Missing)

All values below are **Estimated** for scenario setup and must be labeled “needs local validation” in final material.

| Metric | Conservative baseline range | Why this range is acceptable |

| --- | --- | --- |

| High-risk flag rate among registered ANC pregnancies | 12% - 25% | Typical risk stratification spread in public maternal programs |

| Referral decision-to-departure delay | 60 - 180 minutes | Reflects paperwork + transport coordination friction |

| Referral departure-to-district arrival time | 45 - 180 minutes | Rural distance and transport variability |

| Referrals with complete clinical packet | 40% - 70% | Paper-based handoff incompleteness is common |

| Duplicate test rate after referral | 15% - 35% | Record fragmentation often triggers repeat diagnostics |

| Time from district arrival to specialist review | 45 - 240 minutes | Queueing/triage and bed/workforce constraints |

| Missed follow-up within 14 days post-discharge | 20% - 45% | Continuity leakage without active tracking |

| Manual data-entry lag to central reporting | 2 - 14 days | Register consolidation delays in low-resource settings |

| Avoidable escalation due to late referral | 10% - 25% | Delay sensitivity in obstetric complications |

| Referral status visibility to originating clinic | 20% - 50% real-time visibility | Closed-loop tracking often absent |

D. Citation Rules for Submission

1. Every numeric value must carry one of these labels: ****Observed****, ****Estimated****, or ****Scenario target****.
2. Every ****Observed**** claim must map to a source ID in this ledger.
3. Every ****Estimated**** value must include a one-line rationale and “needs local validation”.
4. Every ****Scenario target**** must be explicitly tied to one or more redesign interventions.

reflection-script.md

2-3 Minute Reflection Script

Hi, I'm [Your Name], and this challenge helped me think like a digital health systems strategist.

The most important thing I learned was that healthcare improvement is not just about having good clinical knowledge — it is about making the whole system work better. For my project, I focused on the high-risk pregnancy referral and continuity workflow in the public health system. That meant looking at how a patient moves from a rural clinic to a district hospital and then back into follow-up care.

What stood out most was how many problems come from handoff failures: incomplete referral information, transport delays, duplicate registration, and missed follow-up. A small delay or missing detail can create a much bigger problem later. That made me realize how powerful digital workflow design can be when it is used to support clinicians, not replace them.

I also learned the value of using credible data and clear assumptions. Instead of making vague claims, I had to separate observed evidence, estimated ranges, and future targets. That made the analysis more realistic and more trustworthy.

Going forward, I can use this way of thinking in real healthcare or operations settings — by mapping the workflow, identifying bottlenecks, and designing practical digital solutions that improve both efficiency and patient safety.

This project showed me that good digital health is not just technology. It is coordination, clarity, and better decision-making for real people.